

Research Synthesis: Evolution of Affective Features in Automated Violence Detection

Submitted by: Meraj Alam

Roll No: U23AI094

Roll No: U23AI089

Roll No: U23AI096

Roll No: U23AI077

Roll No: U23AI075

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat

1. Research Objectives and Core Inquiry

We are living in a time where machines are watching us more and more. This is changing how we think about keeping people safe. Before we mostly looked at where people were moving. Now we want to know how people are feeling before they do something

For a time computers have been able to see what people are doing like running or falling.. This study is looking at why people do things like why someone gets angry and then does something violent. We think that when people get angry they often do things that hurt others.

This study is trying to figure out if we can make computers better at seeing when someone is going to be violent. We want to do this by looking at how people move their bodies when they're angry like what they do, with their shoulders and arms. We know from psychology that people show anger in their bodies so we are trying to teach computers to see these signs.

By doing this we hope to make computers that can see when someone is being abused even if it is not a fight. We want to help computers understand people better so they can keep us safer.

The study is looking at how people move their body like their shoulders and arms when they are angry. We are trying to turn what we know about psychology into something that computers can understand. This way we can use computers to see bad things happening even when they are not really obvious.

2. Strategic Significance of the Study

The more people use surveillance systems the more they need ways to watch over places where people are especially vulnerable. These places include hospitals, places where older

people live, schools, jails and places where people with health issues stay. In these places there are often times when someone acts violently. No one is around to see it.

Traditional systems for finding violence usually look for movements that happen during a fight. Because of this they might not catch acts of aggression, like pushing, hitting or poking.

By looking at the feelings shown through the way people move their bodies the new method helps find these smaller acts of violence. This helps make surveillance and gives a chance to act sooner which makes everyone safer.

3. Application within the Indian Context

Although the research was done using resources from Universiti Teknologi Malaysia the way it was done can be used in the surveillance system.

A real example is in metro railway stations like the Delhi Metro or Mumbai Metro, where many people are watched at the time using cameras. Old systems that look for movement have a time telling regular crowds from dangerous actions.

Also places like government-run Anganwadis, schools, hospitals and homes, for the elderly can use a system that looks for emotions to detect violence. Of just checking faces or using complicated computer models the system looks at how the upper body moves. This helps spot actions even when the video is not clear. This kind of system can tell security people away when someone is acting angrily so they can step in quickly.

4. Dataset Architecture and Modality

The proposed methodology is evaluated using the publicly available **RWF-2000** dataset, which has become a benchmark dataset for real-world violence detection research.

Dataset Details

- **Dataset Name:** RWF-2000
- **Modality:** RGB surveillance videos
- **Total Samples:** 2,000 videos
- **Violent Videos:** 1,000
- **Non-violent Videos:** 1,000
- **Source:** Real-life surveillance videos collected from YouTube

- **Frames per Video:** 150 frames
- **Training/Test Split:** 80% Training and 20% Testing

The balanced dataset helps us evaluate the proposed model in a way. It shows the proposed model what things are like in life. The proposed model has to deal with things like videos that're not very clear things that block our view and places with a lot of people. This is what the proposed model will see when it is used to watch people in a setting. The proposed model has to work with the proposed model, in these situations.

5. Technical Methodology: Features and Modeling

The framework we are using is a mix of two things: learning to get features and classical machine learning to classify them.

Feature Extraction

We use the OpenPose toolbox to get the keypoints from videos. These keypoints help us make three kinds of features that show how people feel.

1. Joint Location Features

We track the elbows and wrists in each video frame to see how the upper body moves. We look at the x and y coordinates to do this.

2. Joint Angle Features

We calculate the angles of the shoulders and elbows using math. This helps us see if someones arms are bent or straight which can be a sign of aggression.

3. Temporal Patch Features

We look at how things move over time by calculating the distance, between frames. We do this over a series of 20 frames to see if someone is making jerky movements, which can be a sign of anger.

Classification Model

After extracting features, multiple classifiers were evaluated using the MATLAB Classification Learner App.

The **Fine K-Nearest Neighbors (Fine KNN)** classifier achieved the best overall performance due to its high accuracy and extremely low training time.

An alternative model could be the **Support Vector Machine (SVM)**, which has been widely used for violence detection because of its ability to classify complex feature spaces. Deep learning models such as 3D CNNs can also be used; however, they generally require significantly more computational resources and longer training times.

6. Performance Evaluation and Results Analysis

Table 1: Performance Comparison

Method	Training Accuracy	Training Time	Testing Accuracy
Joint Locations	98.6%	5.86 s	71.2%
Joint Angles	93.5%	12.74 s	65.5%
Temporal Patches	97.0%	5.41 s	67.0%
Flow Gated Network	—	—	85.75%

The new way of doing things worked well when we were training it and the Joint Location features were correct about 98.6 percent of the time.. When we tested it the accuracy went down to 71.2 percent, which means it does not work as well in the real world.

The main reason for this decrease is that a lot of surveillance videos that’re not violent do not have many people in them or sometimes they do not have any people at all. The method we are using needs to be able to see people to work properly because it uses OpenPose to extract the skeleton of the person. When there are no people it does not have information to make a good decision. Also when the video quality is not good and people are blocked from view it makes it harder to figure out the points, on the body, which leads to mistakes in classification.

7. Conclusion

This study shows that the way people move their body can tell us a lot about when violence is happening. The system we made combines two things: OpenPose to see how people are standing and Fine KNN to figure out what is going on. This works well and does not use up a lot of computer power.

Even though it works pretty well it is not as good, as some other systems that use really smart computer programs to look at videos. Sometimes these videos are hard to see. The system gets confused. To make it better we want to add some things that can help it understand what is going on like how much something is moving and what it looked like a little while ago. This will help it work when the video is not very clear and we cannot see what people are doing.

References

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